PlutoSky R2 SDR Specifications

📌 Product Overview

The PlutoSky R2 is a high-performance Software Defined Radio (SDR) based on the ADALM-Pluto architecture. It features a Xilinx Zynq SoC combining a dual-core ARM Cortex-A9 processor with FPGA programmable logic, making it highly versatile for SDR applications.

🚀 Main Hardware Updates (vs. Previous Versions)

1. **Upgraded Main Chip**: Replaced with **XC7Z020-2CLG484I** (formerly XC7Z020-2CLG400I). While retaining the same Z-7020 core resources, it offers significantly more usable external I/O.
2. **10MHz Reference Clock**: Added a 10MHz reference clock input to lock and calibrate the onboard crystal oscillator.
3. **External 40MHz Clock Support**:
    - *Flash Firmware Mode*: Can use an external 40MHz clock by short-circuiting a specific area with a 2.54mm jumper cap (must be done while powered off).
    - *SD Card Firmware Mode (e.g., Tezuka)*: Users can use GPIO to control whether to use the external 40MHz clock or the internal onboard 40MHz clock.
4. **Optimized Network Port**: The Ethernet port has been moved to the **PL end** (FPGA programmable logic) instead of the traditional PS end (Processor System). This allows customers to directly use logical resources for data transmission without ARM core involvement, achieving true data transparency.

📊 Hardware Specifications

Variant Options

SKU Code	RFIC	Power Amplifier (PA)	LO Frequency	Bandwidth
PLUTOSKY-R2-9361-NOPA	AD9361BBCZ	No	70 MHz - 6 GHz	200 kHz - 56 MHz
PLUTOSKY-R2-9363-NOPA	AD9363BBCZ	No	70 MHz - 6 GHz *(with hack)*	200 kHz - 20 MHz
PLUTOSKY-R2-9361-PA	AD9361BBCZ	Yes (10dB gain, Bypassable)	70 MHz - 6 GHz	200 kHz - 56 MHz
PLUTOSKY-R2-9363-PA	AD9363BBCZ	Yes (10dB gain, Bypassable)	70 MHz - 6 GHz *(with hack)*	200 kHz - 20 MHz

Core System Specifications

Component	Specification
Processor (SoC)	Xilinx Zynq XC7Z020-2CLG484I (Dual-core ARM Cortex-A9 + FPGA)
RF Interface	2T2R (Full-duplex)
Memory (DDR3)	1 GB

Storage (FLASH)	32 MB
Extended I/O	19 × 1.8V I/O pins

Connectivity & Interfaces

Interface	Quantity / Type
Ethernet	1 × 1000/100/10M ETH (Connected to PL end)
USB	1 × USB 2.0 (Type-C)
UART + JTAG	1 × (Type-C)
SD Card Slot	1 × (For PS Boot)

Physical & Power

Feature	Specification
Dimensions	96 × 64 mm (CNC machined)
Power Supply	USB, JTAG, or Pin header

⏱ Clock & Timing Specifications

- **Default Clock**: Onboard VCTCXO.
- **Alternative Clock**: IPEX connector option available.
- **Clock Calibration**: Supports calibrating the onboard clock with a **10MHz external input**. *(Note: PPS / Pulse Per Second input calibration is **not** supported).*
- **Clock Switching**: Supports switching between internal onboard 40MHz clock and external 40MHz clock via GPIO settings or jumper caps (depending on firmware boot mode).

- **Internal IPEX Connectors**: The board features three internal IPEX connectors for:
    1. 40MHz clock input
    2. TX local oscillator
    3. RX local oscillator

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i Frequently Asked Questions (FAQs) Summary

- **Does it support PPS input?** No, PPS calibration is not supported, but 10MHz reference input calibration is supported.
- **Can I switch between internal and external clocks?** Yes, the PlutoSky R2 supports GPIO settings (or jumper caps, depending on firmware) to determine whether to use the onboard 40MHz clock or an external 40MHz clock.